

Introduction (Lecture) ~ Avel Guénin-Carlut ~ Active Inference for the Social Sciences

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All right. Hello and welcome everyone. It is July 7th, 2023, and we are in the first introductory session of the course Active Inference for the Social Sciences. This is going to be a really exciting course and journey that we're all going to join and co-create together. With no further ado, I will pass it to... Please mute. Turn off the live stream, Avel. Thank you. Complex systems and feedbacks with cyber-physical systems will be a theme. We told you it would be. I will pass now to Avel for the introductory session. If you're watching live, please feel free to write any comments or questions in the chat and I'll relay them at the end. On to the lecture. Thanks, Avel.

Go for it.

Go for it.

So, hi. Do you hear me well? There was a sound issue. Okay, cool.

I'm Avel. I organized, I guess, the course. I had the idea and worked on the curriculum.

Do you see the slides well?

Looks good. And just make sure to close the YouTube video or mute it.

So, hi. Do you hear me well? There was a sound issue. Okay, cool.

Okay, I did not know how to do that.

So, this is the introduction session of the course, very unsurprisingly. My name is Avel-Gena Kerlue. I am a founding member of Kairos Research and I am a non-funding member of Active Inference Lab, which co-organizes the course.

And first, I will basically present the teaching team.

So, we have Danielle, which is here with us, which is a PhD in Ecology and Evolution, studying Ant Leavior in an eco-evo kind of way, who is a president and co-founder of the Active Inference Institute

and who is interested in among other things about philosophy and cognitive security.

Then we have Ben, which is also here with us, which is a second year PhD candidate in philosophy at the University of Sussex, which, like me, interested in 4e cognition, which is embodied inactive, embedded, and extended cognition. We don't need to get into that now. He will explain to you.

And in the Proactive Processing, which is basically working on the cognitive science of technology.

And then we have the people who are not here today, which are Mau Albarassa, PhD candidate in the Cognitive Computing at University du Québec, Montréal, which is Head of Product Innovation and Research at

Versus Private Sector Research Lab in AI and the Mathematics of AI, which specializes in Well-Active Inference.

This theory we are working on right now. Script theory, which she will extend. Maybe I will talk a bit of it today, but she will be the one who explains what script theory is to you. And Modern Artificial Intelligence.

Hello, we have Laurence González, also a PhD candidate, also in philosophy, but in Wollongong, I believe this

is how you say it, in Australia, which is centered in philosophy of mind and in the status of information processing, as the term is used in cognitive science. And then there is me. So I made a career timeline, because when I talk about what I did, people get confused. I did physics, math, I did engineering, I did cognitive science, I did research in ecology and evolution, I did a master's in complex system physics, and I did like one year, apparently, given my timeline, or two, no, more than two, of independent research as a part of Kairos Research and as part of the French Society of Cliodynamics. Before I started the PhD I'm doing now, which is something about the relation between mind and material culture, we will get into that at some point. And the question and motivation I have are essentially describing the relation between phenomenology, so the way to experience the world and physical reality, which I have the fringe view that somehow phenomenology is constructive of physical reality, which could have weak and strong interpretation, I subscribe to both.

I'm trying maybe more importantly here to explain or at least describe how social cultural life form, to say the irregularities that in the life of human social life are constructed, which is hard.

And I'm trying to build a research that is effective at doing, you know, knowledge, not word piles, through our work in Kairos Research. So these are specific reference, you will get the slides, and at some point, I guess I will publish them on the website. And then we will be able to check it out, if this is of interest to you. So let us get on the actual relevant content. So what we are going for here is basically explain the very basic demarche of the course, explain how it's of relevance, how the framework we propose is of

relevance to social science, and present a few case studies, you know, make the case. And then we will talk about course layout, but this is pretty quick.

So first I'd like to talk about the explanation of cultural dynamics. What does explaining cultural dynamics even mean?

It depends who talks. There are many, many, many, many, many, many types or views on what is scientific explanation.

I say we can cluster them in three broad categories. One is a nomological, so it's about laws. And what it does is to take

regularities, just say, here is this regularity. And consider that the statements, so the law, has an explanatory capacity toward the regularity.

So for example, for those of you who are familiar with the work of, and I forget names, foundation, the foundation trilogy, I believe it's a trilogy. This is pretty close to what they call psycho history. So basically trying to do, trying to find laws of behavior and using them to predict history in a regular, lawful way. Then we have mechanical explanations, which are more typical of the life science. And what we do in mechanical explanation is to look at the structure of a system, and model the structure, and try to use this model to, you know, predict specific regularities that are of interest. And you have an approach to social sense that is, I'd say, quite close to that, which is cloud dynamics, which weights a lot on the nomological side, but still is centered on dynamics and mechanisms and

explanation by calling on to structure in specific societies. And then you have a functional explanation, which asks what constraints exist so that the system does what it is that the system does. Here again, you will also use a lot of contextual modeling, very often evolutionary game theory. So simulation where we suppose that agents can have specific traits and those that reproduce more, or that have more fitness, you know, give their traits around. And you have the discipline, the study of cultural evolution that does that very concretely by just asking, you know, how culture evolves so what cultural traits can get more fitness and be transmitted, a very evolutionary centered approach to social change. So to sum all of this up, basically, we explain things by calling on to regularities, always. And we can talk about regularities in the phenomenology, so what the system does in itself in the structure that is in the language system activity, or in the function that is the constraints that exist over what the system does. And you have a specific brand of explanatory strategies that try to legislate behavior to make nomological lawful explanation of human behavior and of social cultural change. And it's not immediately evident how they do that, how they think it's a coherent idea to begin with, because, well, humans learn, culture change, so what laws could they be in a field where, which is changing with time always. So in my opinion, it is more a question of, you know, rhetorics than of epistemology, because I do not think the answer is coherent. The answer is, as far as I'm concerned, that people, they just reify regularities that are observed in experimental settings, often that are quite heavily fabricated as kind of cognitive laws of behavior. And because those people usually agree in spirit to the mechanical kind of explanation, they talk about cognitive mechanism or mental processes, but only in so far that those mechanisms or processes are explanatory of cognitive laws. They do not do the work of, you know, looking at the structure and function of the systems. They just claim there must be a mechanism that explains the vague law that I have observed over 20 people that came to my lab. And these laws, they are assumed to scale for the explanation of social activity. And so you have it, you have the recipe for universal law of social change. And then, basically, the whole argument stands on the idea that you have a presentation or mental processes or a cognitive mechanism that somehow entail that the subject's behavior is regular through time and between individuals. And you never have the work that would actually, in any capacity, prove it for reasons that we will not get into because it would get math heavy and ranty. So if the rules and regularities of human behavior can change through learning, through cultural evolution, through what not, how can we exactly model or explain or even describe this agency?

This is not doable in an homological slash lawful framework, because when we do natural laws, we look at regularities, we claim here is a law, this regularity is a law. So it explains that, you know, it exists. And this is a circle because as far as I'm concerned, it's a circular reasoning, like the regularities explain the regularities, which explain the regularities. There is no way out of this, you can recurse this all day long, this, the only fact that you observe the thing is re-explanation for the thing. This is not, hopefully, how we do things in science. And you have a process of explanation that is more complete if you look at structure and function written large. You can look at a specific system, which has specific structure, and you can look at what the system does, or the function of the

system, and at a specific point in time. And you can look at how this function, this activity, shapes the system, later on, at t plus one, you can have, you know, developmental approach to things that is centered on how systems unfold, how they learn, how they evolve through time, like that is doable. There is no law against that, and it could actually work. So yeah, let's do that. Let's explain natural theorem by describing how the underlying structure work, but also by trying to explain how it's brought about, how the system's activity explains the way it takes shape through time. So a claim that we will make here, that is itself, it could be contentious, but we don't have to look at why it is not relevant here, is that active inference, which is a specific new trendy theory of human cognition, can actually explain how regularities underlying social activity be constructed through time. And let us look at what active inference says, precisely. So there is, we won't look at the math ever, we won't look here at the specifics of the theory. The only thing we are going for is to prove that this theory indeed does what I think it does, which is explain or at least describe in a non-tribunal way, the evolution of social regularities. So if active inference framework is true, then cognitive agents perceive the world through expectation they have developed about the world. So perception is an active process. I do not perceive things with my eyes and compute higher and higher level of traits of what I saw. I basically expect things in front of me. And I will actively look at the things that are most likely to inform me. And something that is critical about this framework is that the prediction is also underlies action or policies. So let's say trajectory of action that I enact. I will do the action that I believe I am more likely to do. I will do that in virtue of me believing I am more likely to do this action. Oops. So in that framework, active inference is understood as a process of an action of a specific model that is causally shaped both by an agent dynamics and environment dynamics. And this relation, well, brings a model that corresponds in abstract way to the structure of the dynamics of the dynamics of the relation. And this model, I can also look at it in a way more concrete sense, in that it is what in the lies, what drives the agent dynamic. You have a duality between the dynamics and the epistemic mental states that we put, that we project, that we can observe in the dynamic. So a very important aspect of this is that it dissolves entirely agency, as we would understand the term. There is no distinct step of perception, which informs the agent on what action, on what the world is like. And based on which the agents decide freely what action it will take, there is a constant prediction of what is going to happen. And this is enacted by the agent both through perception and action. So the what drives decision here, what drives generally the flow that drives that analyze the agent behavior is the agent's self model. And that is, that has interesting consequences. So I have notifications, so we'll cut them. So we'll cut them.

What's, let me drink, I will drink, I need to drink.

So one consequence of this, for formalism is that normativity, so the values and norms,

it's hard to go around that world, that drive behavior, they are not, you know, in the brain, they are not something that the agent abstractly relates to. They are embedded in the very process in which the agent understands and gorges and navigates the world. And this is a very relevant for human cultural evolution, because, well, we are cultural creatures, we learn expectation about each other, both explicitly by, you know, looking at how people do things and then expecting them to do the same things again, and implicitly through the very process by which we relate to other people, like norms, like language. And the way you look at this, this, this learning through emotion, it can be, it can be reduced essentially, because of the properties of active inference to what people call regimes of attention. So the way I pay attention to things, the specific things I will look at in my process of understanding and navigating the world, the flow by which I mean that is the same thing as a set of value and a set of expectations. So what is important here is that normativity and values and things like that, they are embedded in my cognition in a much more basic way than, you know, conscious self-reflection. Although, of course, conscious self-reflection can inform that flow. It is embedded in the shared expectation I have with other people. And there is no warranty, no reason that I can actually be reflexive about it. Because, well, the way I predict things, it's based on the statistical regularities that I have seen in the world. And because the way I pay attention, it's, you know, entailed by a predictive model of what is going on around me. I'm very constrained, very much constrained by the statistics I have of the thing I've seen around me. So regularities drive how I pay attention, which drive implicitly, explicitly value and normativity. And you can add the chemical steps, you can add reflexivity, you can add, I don't know, accountability, you can add a lot of things, but you cannot remove the very basic way in which cultural values and cultural expectations are embedded in perception. So, yes, this is a repetition of what I said. The patterns in the flow of attention that are embedded, that I have developed through the very fact of engaging with a cultural niche, and they constrain my actions towards specific outcomes, they entail specific values, specific norms, and they entail, at least they constrain very, very, very, very heavily the intention that they're even able to conceive, to generate.

So, there is a dual, there is an equivalent way to see this, which is engagement with the cultural landscape. So every agent, they experience the world, not in the same way, of course, but they experience the same. Let us say that each person's experience of the world, it predicts very strongly the experience of the world that other people will develop around them. So there are structural properties of the, you know, landscape of opportunities and perception and possible action that we experience.

There are properties that are transferred between individuals robustly. This is the case, for example, of the norm. So if I say, if I say, if I look at a law, any specific law, all of the people in the same country, they are subject to the same law. So maybe some of them can just ignore it, they live too far from the city, we don't have cops here. Maybe some of them are above the law in some capacity with their relation. But the basic constraints that the law entail, it is experienced by everyone in a different way. We will all expect that specific people will react to specific situations in a specific way.

And this is a special property of the landscape that we experience. So the way that this landscape shapes exploitation in a robust way, we can look at basically it's as a constraint. So not as a derivative property of a landscape that I can or cannot experience, but just as a statistical force that shapes behavior in a specific way. And looking at things like that, it means that I can see a given social cultural field in two complementary ways. As a set of agents that navigate a common, well, not quite common, but mutually constrained social cultural worlds, or as a set of constraints that produce itself. And the way agents reconstruct the world around them, in their mind, the way they understand it in a creative manner, it is equivalent to the way the set of constraints reconstruct itself. Which is important because if I can reconstruct, if I can look at this world as a set of constraints, over statistical constraints or behavior that reproduce itself, I can look at it as a living thing, strictly speaking, because this is how we define living things, at least in specific sub-branch of system biology that I believe is relevant, is to be believed about that. So when we talk about structural landscape, we can look at them either as a landscape of affordances, this is an opportunity for action that are experienced and enacted by a set of individual agents, which interact so as to having some level of coherence between the landscapes they experience and enact. And I can look at the same landscape as a system of constraints that are exerted over the behavior of those agents. And that I can define and study with no clear reference to any individual agent in the same way that I can study a dog without looking at every single individual cell. What you're interested in biology or sociology is usually, you know, you know, aggregated patterns, not what every single living thing around thinks of every single thing.

So that was the heaviest it got, probably over the world course. I do not expect you to grasp all of the consequences of what I said because it was very condensed and it was very abstract. But the case I'm making here is not that everyone has to understand all the math right now. It is that this framework, it does things to say about relevant questions in social science. And to make this point, I will not just, you know, give the abstract presentation of how it's awesome because we have landscape, whatnot. I will look at specific fields, questions, case studies in social sense. And I will look at basically how active inference entails a pretty specific model of that. That is well grounded in a cognitive sense and that is relevant and that is adequate to the study of living, changing systems, etc, etc. So, first, agenda setting. I think it's something that is situated in the work on public opinion, specifically, like the, I believe this book is called public opinion.

So, agenda setting, it refers historically, also, you have a more diverse way to understand it now, to the role of mass media, like journals, and later TV, in manufacturing consent.

This is a textual station from the text of, no, I thought I had the book cover. I forgot the name of the guy. I'm sorry, you have to look at the reference.

But it is used to study how mass media can manufacture consent by drawing attention to specific issues values and shaping specific expectations about the world. So, the idea is that you have a role, an active role of the elite, what I call the elite here.

People which have outsized influence of the flow of policy or society. I will call them elite.

And so, those people, they basically transmit norms, they transmit normative expectation, so, concerns to whatever is the thing that they do, that they want you to do, more specifically.

And they do so not by telling you, do this and that, they do so by building agendas, so, drawing attention to specific issues in specific way.

And you can see here, people who are, I mean, it's prevalent in the political life, everyone does that. But, the US right wing is very, very, very transparent in doing that.

And so, this is a meeting between two experts in agenda setting, Donald Trump and Ron DeSantis, which basically have convinced people that

that they should not worry about the price of life, whether we wage war here and there, why do we have to pay millions, why do US people have to pay millions to get healthcare, but to how LGBTQ, democratic people threaten the US life, they threaten the US life in a very abstract way. Like, I did not ever hear any consistent coherent believable mechanism by which they explained how LGBTQ threatens whatever Republicans.

But they do draw attention to this topic. They do shape the expectation that Democrats and LGBTQ are predators.

And they do so in a way that is able to legitimize themselves as a beacon of the people, to mobilize people to actually, you know, be behind them, and do their bidding, and to canalize political life in a way that is convenient for them, because it, you know, keeps them in power.

And the role of attention, the way you pay attention to things in shaping behavior, it's just a redo of what I explained in active inference.

If you believe that the way we, the forces, the way we constrain attention is explicative of behavior, then you believe in active inference, just express differently.

So this is a case where you have classical, let us say, concept in a certain sense that essentially reduces the concepts of active inference.

It's not to say, of course, that astrology is irrelevant and we should just do cognitive sense, but this is pretty rare to have such a direct response between topics that are disciplinary this far apart.

So this is something we should definitely pay attention to.

So second case theory, nationalism.

Basically, this case theory is entailed by what I said about the Republicans, because we move by which they legitimize their rule.

It is just say, we have, we share an identity here, US conservative, traditional values, etc. etc.

And this identity, it entails specific norms.

And for reasons that we are usually left out of the frame, I, so elite people, Renaissance, Trump, Napoleon, whoever, I get to tell you what this identity entails in terms of behavior.

You don't get to speak, you don't get to speak, because you are not someone who has newspapers that parallel to what they say, what you say, because you have to keep on the subject.

And this is very consistent with the dynamics by which we constructed nation states.

So for the historical context, the idea that we have ethnicities, and this is somehow relevant normatively, like the Slavs should have organized together or the, I don't know, Hungarians should have organized together in the

We should get to agree to things that we do, and then do them. It's very recent. It's extremely recent dates.

Well, I won't get into the specifics, but let us say in this specific formulation, it dates from the 19th century, when you had a bunch of very powerful

social administration that took hold respectively of France, United Kingdom, Prussia, and they had the capability to rule most of the social activity.

But they did not have any clear argument why people should, you know, listen to them in any capacity.

And the way it worked is that those administration, they basically started claiming that they were some representative of a given

ethnical and linguistic community. And that in virtue of being part of this community, you should listen to them and pay them respect and actively work for their goal to be fulfilled.

And of course, it is not that easy to make this claim. But what happened is they invested a huge deal of energy and organization into actually having people buy that they built schools,

they built prisons, they built psychiatric hospitals, they built a lot of infrastructure that enabled them to transmit norms in a very robust way and transmit identity in a very robust way.

And in doing that, they managed to embed most of the micro social relation you could have into their specific ethnicity and into the normative power they lend to this ethnicity.

So at some point you were, you know, I always forget, farmer in a lost village in France, you only meet the situation at tax time.

So main arm come and tell you give your grain and give your grain because else you're dead.

And at some point, those people send you a teacher and they say newspaper and they claim, hey, you are French, you have to speak French, you have to be friends with us, you have to fight our wars.

And because there is so much energy that is spending that people actually bite or die.

That's another option because, you know, the other side of the coin is that states that legitimize themselves through a nationhood, they have basically to be all the time invested in ethnical cleansing.

Because people who are not considered elements of the state, they have to be submitted or removed, which is a bad thing to do.

This is a very bad thing to do.

And I'd like to illustrate this whole notion of identity capture through the African Renaissance monument that Abdoulaye Wade, the president of Senegal,

was constructed in the capital.

So it's very glorious.

It is very good example of identity capture because it shows you the norms.

It shows you a glorious image of the African man.

And it shows how you have to, you know, be supportive of children, show them, protect them with your big muscle and give them very well ambition.

While African women look at you like they were non-agents, let us say that.

So it gives a blueprint around which society should be organized.

It gives you a very specific picture of who this is speaking to, in this case Africans, because I guess Senegal gets to say what is good for Africa.

That is coherent claim to make.

I mean sarcastic, sorry.

And why I like this specific monument, well, you have a lot, lot, lot.

Well, first it's very expressive.

You don't have the Arc de Triomphe in France.

It serves the basic same function, but it does not show you, it tells you.

And the African Renaissance monument, it shows you what it means in a very, well, obvious way.

And the second main reason is this.

You have nothing around it.

They just put it in a pile of dirt and they invested nothing in the infrastructure because they were too busy, you know, keeping, taking money for themselves, I guess.

I don't know exactly what I did with tax money, but clearly it was more invested in the symbolism of what African men should do and why they should follow him specifically and not in concrete infrastructure.

And in most cases, identity capture is a very raw deal for people who are, you know, captured.

This is why I call that capture because what you get out of identity capture is a symbolic sense of being part of the good team.

And what you have to give is usually, you know, the entirety of the work you do over your life.

It gets given to the state or administration that wants you to do things and you do things for them.

And sometimes they build infrastructure, but if they don't want to, they will not.

And that's, that is very, very, very, very much a bad deal.

And I, I give it away.

So national identity, it was diffused to school and mass media.

It was embedded in micro networks of solidarity.

So typically the African Renaissance monument gives you, basically tells you that as a husband and father, you have to give in whatever the state ideology of what was.

And then it is leverage for mobilization and legitimation and control.

And you have another, that's, that's not, oh, I did not say why it was relevant to active inference.

Well, it is relevant to active inference because all of this is mediated by self model.

All of this is enabled because people have self model and they enact the self model.

And there is no necessity of a reflexive step where you actually, you know, think about what is your self model.

And you are expected to do a thing.

You do the thing because you expect yourself to be doing the thing.

And active inference, as we said, you do what you expect to do.

So this basically explains why such symbolic, you know, raw power, it has actual consequences.

While if you have a, you know, more rational model of humanity, the question is, hey, why you build big things and not infrastructure?

It would be a rational response.

But this is not what people do.

People follow norms.

And this is a way to broadcast norms very efficiently and robustly.

So due to that, I'd like to talk about the structure and agency.

So there is a constant question in a social sense.

Usually social sense talk about, you know, structure.

So norms, habitus, way that institution builds life.

And there is a legitimate question that is asked, which is, if you believe structure, treatment and behavior, what is the pace for agency?

So there is, there are constant debates in every field possible of social sense regarding what is actually explained by the structure and what is left to individual agency.

And what I want to say is that this question is essentially dissolved by the model of agency we have.

So, as we said, agency is exerted through expectation over myself.

So because social structure, it becomes expected, it is enacted through the agency of the specific people that are within it, not have an opposition between how strong institutions are and how agentee people are.

Because strong institutions are institutions that basically become embedded in your self model.

And so are, you know, there are people freely, not freely, it's a bad term, but from their own accord, they enact those institutions.

And this is the whole thing about an anti-capture which I presented.

Nationalistic ideology is strong, is efficient in building actual mobilization, because it makes you believe that you will actually, if you want to court, follow the, uh,

the, uh, social order it, uh, gives, which was, which was not represented, but pretty rare in a pre-modern institution.

Usually, for the institution are, um, I am the king, I have an army, give me food.

And when you have legitimation, uh, attempts to actually explain why I should get the food.

Um, these are pretty much justification.

This is the, uh, elite people talking to themselves about what, why they are justified doing that.

But you have no arguments and you have, that would be actually compelling so as to why you would have to give the food.

And you have no attempt whatsoever to, uh, convince people who have to give the food, they should give the food.

This ideology, it's not, they do not talk to the people they ask things from.

They talk to themselves about why they have, they can't, they are legitimate to take the things.

And so, yeah, in, in those, uh, case, you have a pretty evident conflict between structure and agency.

Do I, uh, do what I'm told to do under threat of death or do I do any other thing?

Of course, most people would like to do other thing, but they can't.

And they will feel, or they can't.

But if you accept that agency is built through normativity, through engagement with a social cultural field of finances of party for action.

Yeah.

You have to admit that structure exists at least in part through agency, because what agent believe is

legitimate, what agents, uh, believe is right.

Right. What agent believe is expected.

It is very precisely what constitutes structures.

And that is, uh, insight that is us at least in part left out in of the conversation in, in some field.

And like to, um, to show why it's, um, the direct conflict between certain agencies, it does not have a strong grounding.

Uh, yes, again, I have said the thing that the slide would say.

There is another thing that is maybe closer to my core body of work, which is, uh, how active inference afford indirectly an archeology of cognitive landscape.

So, um, if social organization is made of constraint or behavior, uh, and those constraints are to some extent embedded in the natural landscape,

in the natural environment of the people, then we can, to some extent, study archeological, archeological landscape for themselves.

We can look at the landscape, like material landscape, the, um, objective study of archeology.

And we can make some inference over the constraints that this landscape entails over collective behavior.

Uh, I will not get into details as to why, because it's, um, it's, uh, far call.

Making this inference is difficult, but very concretely, we can ask how, whether, uh, hierarchy.

So that is that some people get to say to other people what to do and certification, the idea that some people are by nature above other people, uh, that can be embedded in vertical patterning.

So you have a claim by people who found me escape project.

Look it up.

It's nice.

Um, you have a claim that to some extent, the rockification of society around the iron age, it is written.

It is embedded in a vertical patterns in material culture and architecture.

And maybe I will not buy the claim outright, but it is very compelling.

If you look at specific architecture, you'll get a specific feeling.

So this is the, um, not cathedral, but maybe basilic or friccatic.

Uh, I will not try to say the name because I do not speak, uh, Icelandic and that would be bad, but you have very progressive, compelling curve towards the sky.

You have specific feelings, uh, that are, um, how do you say produced by this architecture.

And you have a very specific way in which it drives attention from some, something wide on the ground to something that goes up.

And if I did not make it clear yet, if we buy it in France, we buy that, uh, the patterning of attention, uh, the way you give attention to things.

It carries norms, it carries intentionality.

And in first person by introspection, you can, I think all of you can very clearly see how architecture drives specific patterns of attention and how those patterns specific, elicit specific feelings.

And we currently do not have models that would be very specific and very predictive of how this work, but we could.

This is, this is huge work, but this is doable work.

This is not speculations.

And I do not know of another formalism, an active inference that would actually make it possible.

And last, I'd like to, uh, talk about, um, uh, the favorite work, my favorite thing in social science, I guess, which is, uh, sing like a state, uh, Scots anthropology of states.

So the core argument of Scots, I'd say it's hard to identify a core argument in that book, but you have ways in which states.

So here equivalent to administrations, they understand the world and some world through, you know, written forms that they can pass around, uh, many people, many institution to have actually a coherent model of the world.

That is, that is pretty much obvious.

And what should be obvious, but is not, is that those models are reductive in the sense, any model are, they emphasize specific information and they leave out other specific information.

Uh, something that is very interesting, I'd say is that states, they tend to push specific models, specific landscaping, specific norms in urbanism, specific, uh, family norms that build legibility.

So the ability to be understood by the state.

So for example, uh, you could have, you could have very sparse, uh, settlements that are quite autonomous and that do the thing.

It is a possibility.

There is no law or physical constraint around that, but it is far easier as a state to manage what is very close to it.

So for a state, it's good to have, you know, a population that is very close to the center of power because those people, they are just there.

You can go tell them what to do.

It's simple.

And you can, you can go ask them things.

It's simple too.

It's very easy to manage that.

And those people that would need, you know, food to stay there.

So we need some kind of food production model that is easy to take from, and that is easy to organize and engage as a centralized institution.

And this is monoculture.

This is agriculture as we understand it currently.

This is people that make big fields that make the same crop always cereals.

Well, not always sometimes potatoes, but mostly cereals because cereals, you know, it's very dense.

And it's typically it is easy to keep.

It is easy to transport because it is dense and it is, um, it's stay edible long after, um, you know, um, recalled.

You say recalls English crop cropping.

I don't know what is the word, but when you take the food from the, uh, plants and people have to stay there for very long to cultivate that.

So they cannot, you know, go around and, uh, put potatoes here and there and leave when they don't.

So you can just go to people and you will know where the people are and where you can take the food.

And, uh, you know, when the food is going to be, um, to be, again, I don't know the word cropped because it's real.

It's always the same time.

It's very visible in the landscape, uh, how far you are from, uh, recalling slash cropping.

So you cannot really have a way to cheat that.

You can just send men in arms, take the food, give it to the city to depend population.

And you have a very, um, rationalized, controllable society.

And we, we do not know that this is the reason why we have cities and, uh, monoculture, but we do have traces that states very consistently, uh, try to have big cities and more culture fields.

And mostly they succeeded.

And, oh yeah, this is relevant to active inference because in active inference, the way you pay attention to things, it is productive of your world.

It is productive of what cognition you have.

And here, uh, the capacity for administration, the capacity for written symbols that tell what things are, they are critical to a state's understanding of the world.

They are what literally produces the way they pay attention to things.

And the fact it was, um, basically major thing for them to have a world that looks like this.

It is, uh, very directly telling about, you know, the fact it creates a world.

It's a direct instantiation of that.

And that is a nice link to do.

And I'm happy.

I did this link.

So, uh, this is for the case studies, uh, expectedly, you don't see the corrections, not all of them.

Maybe you're convinced that there is something there, but most likely you don't see very precisely what is to be seen here.

So the expectation is that over the course, you develop a very specific understanding of active inference, the framework.

Uh, you develop a specific understanding of how it relates to social science, to questions that are relevant in social science.

And we develop the capacity to, you know, have an informal opinion.

That's a critical informal opinion of how this relates to question in social science.

So, uh, we are doing the introduction lecture.

Hopefully I've convinced you there is something to see here.

If not too bad.

Uh, and the next session will be in four days, not one week, uh, four days.

It will be about, uh, basically the basics of active inference.

What you have to know to understand, not the framework per se, because it's math heavy, but what the framework says about our life as community agents.

So then you have the discussion.

I, you should know that for each lecture, there is discussion.

Lecture is, uh, you know, vertical people talk, you listen, discussion is more free form.

Uh, every person has a, basically, um, authority on how they share the discussion.

But the expectation is that it is a discussion because it's the name we gave to the thing.

So after the basics, we'll look at, uh, patterns of collective behavior.

So how those elements on the commission, human commission or non-human commission, they give tools to study the organization of collective systems like, and colonies or societies, but we'll stay in, um, not specific to human approach.

We'll be interested in collective behavior, but we're not, uh,

be interested in the specific patterns of collective behavior as afforded by, uh, human commission.

Then we'll look at, uh, semiotics and semantics, the evolution of symbols.

So things that are closer to humans.

We'll look at how active inference accounts for evolution of symbol, and what evolution of symbol affords for commission.

And then from here, we'll look at, uh, very social sensey things, norms, script, narrative language.

So how this discussion on active inference and collective behavior and semiotics, they inform very directly to give a model specific principled predictive model of, um, social human behavior.

And then we'll look at, so my thing, uh, the way I like to think of it, all which is short constraints.

And we'll go back to the claim that there are landscapes of, uh, social cultural commission thingy thing.

And those landscape can be looked at in, uh, you know, abstracted from the individual scale manner.

And this is explanatory of cultural change.

And then, uh, we go for the conclusion.

So we'll be interested in your opinion of the course.

This is the first one.

We'd like to have feedback, a lot of feedback on what you thought of it.

And what would make it good.

Um, so that will be for the, uh, conclusion discussion.

So here, I hoped that you'd present yourself.

Uh, we have not the capacity right now to have, um, students take the screen and talk.

Uh, but I'd like to know, uh, you can say in the chats, uh, what is the background?

What are your expectations for the course?

Whatever you want to say about, you know, things, the world, preferably, um, being related somehow to the

course.

That would be nice.

Uh, but yeah, please tell the things to us.

That concludes my talk.

I thought this light.

Right.

Let it be clear.

Uh, we have, uh, last thing maybe.

We, uh, considered making, uh, projects.

So you making projects that we, uh, read, um, maybe, um, to say, right.

Uh, but I do not think that we are in capacity to give, uh, you know, um, um, standardize normalized degree or attestation that has any value.

So we decided not to do it, but if you, so asking you to do things, uh, to do projects is a bit weird.

If we're not gonna, you know, grade it and attest that you did it.

But if you want to do it, if you want to, uh, write things and send them to us for review, this is, uh, this is very okay.

You can do, you can do that.

You are not formally invited to do that, but you can do it.

Well, I'll just add a point on that.

And then meanwhile, if anybody wants to ask a question in live chat, um, we did raise this topic of different qualifications and it's something that we've been working on and continue to work on in the Institute.

So potentially in future versions, but absolutely in future offerings, we will have these kinds of courses.

But as I've mentioned, this is kind of a first run.

And so it's really exciting just to even start that conversation and see what people are looking for in terms of certification.

Um, all right.

So let's go to some questions in the live chat and comments.

So I'll begin.

Oh yeah.

Go ahead.

Um, I did not make the explanation clear, but you know, because it is a first and because we are still checking the course, it is so much more important than to actually tell us what, what you expect out of it.

Like what you want or what you perceive will be here, you know, expecting the active inference sense.

That is a bit perception, a bit action.

Please tell us about that.

Awesome.

In fact, the only way to reduce one's uncertainty is through active engagement.

And what is literally applying active inference when they do actively engage in this epistemic way.

So I hope that all comes through.

All right.

Um, we'll see a first question from Sfumato in the chat, does active inference provide us with predictive power, potentially focusing on human or social skill issues?

What does it look like, or where would you say active inference provides us with predictive power or unique explanations and predictions?

So, uh, yes, or no, depending on the granularity of the question.

of your question, so what you look at active inference and not active inference.

Active inference is a branch of predictive processing,

so it's a family in cognition that consider that your perception of the world

is based on prediction. You have two specificity of active inference

within predictive processing. One specificity is that it says that

actions are predictive, are based on prediction too. The other is that

it proposes a mathematical formalism in which to study that.

That grounds the claim, but also constrain the mechanisms a little bit.

Predictive processing as a family is predictive of many things.

It's predictive of the fact that basically you think you see things,

but you are not in capacity to perceive them right now.

So I see a bit of tissue, no, fabric that is behind my computer,

but I know and I see the patterns in it, but I know perfectly well

that the cells that are actually, that have the precision to see those patterns,

they are looking at the screen now, so I cannot possibly know what the patterns are,

except if I expect it, and this is what shaped my perception.

So this is predicted by predictive processing, and what is also predicted is that

basically the information flow in the perception part of the brain,

it is heavily top-down. You have more bandwidth that goes from visor cortex

to the retina than reciprocally, which is simply not compatible with

direct perception, passive perception kind of model. It is, however, compatible with the idea that we predict the content of retina activity,

and when our prediction is wrong, then you have feedback. You have perception, error, propagation.

So this is pure predictive processing. This is explained by predictive processing.

This is contradictory actively with most conceivable variants of non-predictive processing.

So yeah, predictive processing is predictive and I'd say it's not up for debate, as I know.

Active inference, it takes in addition the idea that the actions are based on prediction.

I do not know what empirical work existed to check whether this is specifically the case.

Maybe Daniel, you know what has been done about this?

Complete your answer and I'll give a thought.

Okay. So there are conceivable empirical settings that would check that out.

I do not know of them, so I will not claim that they exist or do not exist.

What is important to me is the mathematical grounding because most models of cognition kind of assume that we have computational power that is implemented somehow, somewhere by, you know, meat.

And this is wrong. We cannot just suppose that the big properties that the brain must exist somehow, somewhere.

We have to actually explain it. And this is what the mathematical formalism behind Active inference, the Frenzy Principle, does.

It explains that we should expect the system to be predictive somehow.

And although it is not directly embedded in math, it is pretty easy to see why the brain should have a lot of power at doing that because of the, you know, very heavy tuned power for the mechanical attunement it has.

So, but I would not, if active inference is empirically distinct from other case of predictive processing, I am not sure how I cannot think of an experiment or empirical observation that definitely discriminates between active inference and non active inference predictive processing.

I think it exists.

Great. Thank you. Well, in many ways, active inference is predictive processing with action under additional sophistication and richness supported by the free energy principle.

These are technical topics that we explore in other discussions as to whether active inference gives us unique explanation or prediction capacity.

One could look to, for example, the linear regression or any other type of statistical framework.

Linear regressions in and of themselves don't make predictions because you need a target to make a prediction about.

However, given a linear regression model relating to health outcomes, for example, you might use that model to reduce your uncertainty, which is to say have predictive power about that system of interest.

And so in that respect, active inference has been used to develop a host of unique explanations and predictions related to a variety of cognitive and behavioral phenomena.

I'll just leave it at that.

And I'll read some introductions and such in the chat.

So Robin writes, hi, my name is Robin.

I work in digital governance.

Yeah, go ahead.

What makes that I buy active inference as a mechanical model of human cognition is that you don't really get adaptive, chemical attune, adaptive action without some variance of action being predictive.

Because the default picture would be that you have to planify a full action trajectory, but there is no computable way.

This can be fast enough and robust enough and not costly enough to underline adaptive action.

So you'd be basically, you'd say, I have to grab a cup.

I would look at the cup, see the cup, planify movements and act the movements.

If something happens, if I was slightly wrong about the position of the cup, the algorithm is too heavy for me to adjust.

So you not be able to navigate anything at limited information or this is changing.

And we are very good at navigating things that means information, which are changing.

So until there is another mechanical model that predicts, you know, attuned adaptive action, I will have to buy active inference as a model of human cognition specifically because the mass underlying is pretty solid in my opinion.

Yeah, it's always a great question is relative to what measurements need a comparator.

And so active inference with predictive value relative to something unspecified doesn't really make sense, but we can evaluate it better or worse in a certain setting.

And so it's important to compare it.

So I'll just read a couple introductions.

Robin wrote, hi, my name is Robin.

I work in digital governance.

I've been curious about applying active inference to the construction of governance for digital commons.

Great kickoff.

Sumato wrote, hi, I've been studying fundamental physics, and now I am doing Clio dynamics with a current interest in pragmatist philosophy.

I came to discover active inference and see how to apply it to cultural evolution.

Ander says hello.

Anthony writes.

Bonjour.

Anthony writes.

Background, poker player, father, entrepreneur, very little knowledge on subject.

Attend cognitive security in Davis.

Working on project that active inference is very relevant to.

Expect to learn and engage.

And Gina writes.

Hi, I'm a biologist and neuroscientist that wants to understand the theory to do experiments with the Xscape team.

Cool.

Well, people can continue to add introductions or questions in the chat.

Ben, if you want to ask or reflect on anything.

Or I'm sure there's more things to talk about.

Yeah, I thought that was a fantastic introduction.

Some very interesting connections that I wasn't expecting there.

Nice to see manufacturing consent making an appearance.

And a lot that I think I can.

Hold on, let me turn on my camera.

Yeah, certainly a lot that I want to pick up on and hopefully circle back round to in the in the in the session that I'll be doing.

Cool. Cool.

By the way, what I wanted to say earlier, and I could check now is that I believe I believe that manufacturing consent was a Chomsky thing, but it is not.

It is a Lippmann thing, which is the guy that basically founded the very notion of public opinion in the 30s in the book called Public Opinion.

The book is co-authored with Chomsky there, no?

Chomsky was not.

It was maybe eight at the time.

I very much dubbed he actually did a thing.

I'm concerned.

I'm sure we have a copy of that book on my shelf, and I'm pretty sure it's Chomsky.

There are no copy of the book Manufacturing Consent by Chomsky.

The expression Manufacturing Consent is from a 30s book, Public Opinion, by Walter Lippmann, that founded the whole public opinion thing.

According to Wikipedia, at least I did not look for a specific book that would specifically mobilize public opinion.

Because like, what do we mean by public opinion?

I don't know.

A lot of them go clearly, but maybe something else goes.

I don't know.

It's created for that.

I can at least say it.

That's kind of the interesting question about multiscale systems and emergence.

If we enable or grant that brains can perceive or do things that a single neuron can't, but that they have a material relationship.

To one another, then is public opinion just like a pile of sand?

It just represents the summation of the nest mates?

Or does public opinion represent a singular integrated unit in its own respect, one that might have to be sampled by investigating the material constituents, but still being something different from that?

I'll just ask, go to a question in the chat or Aval if you have a thought.

Does a pile of sand have semantics?

I adopted.

For an observer or for itself?

Sorry?

Does the pile of sand have semantics for an observer or for itself?

Good question.

For an observer, I guess, not really.

For itself, clearly not.

Really not.

That is symmetry.

But I can put together brain and public opinion.

Different coordination mechanisms, but still coordinated systems.

Yeah.

Complex adaptive systems in an unfeeling sense.

If the sand is piling up at the bottom of an hourglass, it certainly has semantics for an observer.

Who knows about hourglasses?

Michael says hello.

Haptivist, greetings Michael.

And asks a question.

Please further contrast the terms regimes of attention and sociocultural landscapes.

Contrast as in explain what the terms are, I guess.

Yeah.

How are those terms similar or different?

I do not see.

Basically, we are, I propose we active inference people.

I did not have the idea.

Propose regimes of attention.

So specific patterns in the way you, you know, your attention flows.

As an explanatory mechanism, let us say, what underlies the experience of attention.

The experience of cognitive landscape.

Cognitive landscape, you understood whatever is the array of things that you as an agent experience as your world.

And then we, active inference people, also claim that the coordination between individuals in, you know, cultural agency.

So the way we co-construct synchronize regime of attention means that there is a coupling between the landscape that we experience.

So landscape that are constrained in such a way, I will call social cultural landscape because it's easier than saying cognitive landscape that are constrained by social cultural processes.

Is that satisfying?

I'll just give another view on that.

A socio cultural landscape is something like the ecological landscape or the actual basis materially of our system of interest, what we are studying in sociology.

And so it consists of inert objects as well as animate cognitive entities.

Whereas regimes of attention refers to the dynamical patterns of attention, which is to say relevance and salience for cognitive agents.

And so the socio cultural landscape, to use an example from Avel's presentation might include the modified niche that includes a monument.

And then a given cognitive agent, when they gaze upon that landscape in a cultured fashion, have a regime

of attention such that their eyes are moved towards the point of a monument.

But the regime of attention is modeled as a pattern of dynamics of a single cognitive agent.

And in active inference, we model attention as covert or internal action.

I want to add something I use possibly abuse the term landscape because it has a very specific meaning slash connotation, which is a space that you can gaze at.

This landscape is a space that you can gaze at.

If you go to a more mathematical notion of space,

you will basically find very closely the expectation

or formal entailments that this space represents the set of possibilities

accessible to a system. This is all there is to it.

This is not what a cognitive landscape is.

It is not the set of possibilities that abstractly an agent could experience.

It is what an agent experiences now, what they can gaze at now.

This is why the notion of a landscape is very heavily used.

It does not have a clear mathematical meaning.

It should have because it has a very specific conceptual meaning

that is critical to the understanding we have of cognition,

and that is not afforded by set theory.

So most mathematics we would use,

the brand of mathematics we teach in school and that physicists use.

Most physicists, not all physicists.

Well, if we consider like Waddington's epigenetic landscape,

or if we consider the surface that variational free energy is optimized on as a landscape,

it gives us a lot of really good intuitions.

It gives us something to gaze upon as instrumental modelers,

but it is not the same as the landscape of trees,

even though each one of those trees might be modeled by an observer

as possessing an epigenetic landscape or a regime of attention or a free energy landscape.

Well, if anyone has other thoughts or questions, we can consider them.

Otherwise, uh, Ben...

Just give me a second.

I don't think everyone has a...

Do you see the image?

Yeah.

It's small, but Waddington's landscape.

There you go. Thank you.

Yeah, Waddington's landscape.

So it is a very old, uh, thirties, twenties, thirties, I guess,

thing in, um, evolutionary biology where, uh, so Waddington is, let us say the founder of the contemporary developmental biology and the landscape are basically a dynamical, uh, fitness.

I do not know how to not say landscape.

That is a...

They're basically the space in which, uh, agents can evolve.

Uh, the phenotype of the agent can evolve during its development.

And, um, this is a very, very, very, um, grounded, uh, visual, um, metaphor, symbolism, I don't know, uh, for the study of the developmental system.

So this is why we talk of landscape for the gazing and for this historical grounding of the developmental, uh, studies in biology.

And also I literally am paid to landscape biologists, uh, landscape archaeology.

So yeah, let's overuse it on landscape.

That is good for me.

Cool.

But it's an example of using our spatial and embodied experience to, to get a grasp on different conceptual topics.

Ben, you have any comments or, or I'd love to hear a little.

I'm gonna actually ask about this, this image.

If it, there's a couple of words that I was expecting to creep into this discussion that, that did not, which are, uh, attractors and repellers.

Ivel, when you mentioned dynamical systems, I wonder if you think these are useful terms for thinking about, cause attract, attraction goes very naturally with thinking about attention.

And I wonder if you think this is a useful, um, kind of linguistic tool for visualizing the relationship between, uh, social cognitive landscapes and, uh, and attention.

Um, do you see the social constraints diagram now?

Okay.

I wanted to have it.

If I, there is a screenshot laying around, I wanted this to be a screenshot.

So attractors and repellers.

Yeah.

I mean, we're basically talking, uh, fitness.

When we're talking landscape, we're talking fitness landscapes.

So I guess it kind of entails the notion of attractors and repellers.

Uh, no, there are many reasons why it stay away from the micro systems terminology, because, well, um, the micro systems, they entail the pre-formulation of all possibilities of the system, which I think is very, very, very wrong to do for biological system and social system system that are, that undergo, uh, upon the

[illegible]

timescale it's okay but then if we have to uh abuse the formalism this much why not not use the formalism that would be my uh flow of reflection about this terminology i understand the tractors as uh states that have visited more regularly than others so is there something about that that's not correct i will check it out yeah an attracting set is one that is revisited but there are of course many sub definitions to be more technical like the attractor is it just the point or is it the basin of attraction but broadly on a landscape as things shake out the attractor begets some states that are revisited more than those that are not that's kind of the tautological definition of the attracting set so so attractor is emphatically not at least under a standard definition uh set of states that is visited more often it is characterized by forward invariance so the fact that once you're in you don't get out so an attractor is a manifold so a subspace let us say uh such that if you're in get out well there is a basin of attraction such that if you get in the basin you get into the attractor uh so under if you look at stochastic chemical systems maybe you will get out because it's you know stochastic means uh with random noise basically let us say that so if you are specifically within a stochastic chemical system so not the mechanical system per se um then you can talk of attractor and think of things that are not actually uh that do not have this forward invariance just things that you tend to stay in for a while once you win but you are not talking about attractor anymore you are talking about some other thing the functional property is foreign invariance we don't have the invariance here so let's not talk of attractors i do not yeah but we can we i mean we shouldn't fall down this rabbit hole completely we can talk about this another time let me go to another great question in the chat michael says a philosophical question science does a beautiful job of describing and building testable predictions of reality but underweight subjective experience human subjectivity arises within experience of observation but includes preconditioning as well as experience how should science informed observers reconcile the limits of subjective experience with the limits of objectification science build good models good models are our competent engagement that is basically it um so the question of how to build a sense that is effective to information is basically the question uh uh that on the kairos i would not use the word effective perceive but you know that's pretty close and we'd have to in principle sense is built to uh create precise models precise i think does not create prediction error within a specific context that is a um heavily heavily ritualistic and conservative and that would be you could criticize pretty heavily the institution of science as a truth seeker normative thing but you so you you'd have to have a good level of reform to basically enforce a modeling that are both robust and easy to transmit socially outside the uh cast of scientists so uh daniel and nine many other people wrote a paper on the ontology for decentralized science which speaks to it and um

yeah so if you get science you get good models and once you are good models you can navigate competently

uh but slash and i don't know but and both what bend bend uh bend um you would need a heavy level of reform institution of science to get this process be more you know ergonomic efficient compelling and uh socially influential and uh yeah i think it's an important thing to do when i try to do it so thank you for the question i'll give another angle on that bayesian statistics

qualitatively equates to the explicit provisioning of our priors and how we update those prior beliefs as new observations come in we may have high attention to new observations and update our beliefs a lot we may not be paying to the attention to the new stimuli and therefore continue our prior belief so if we're not paying attention it's kind of like a belief with inertia that's greater and if we are paying a lot of attention then beliefs change more rapidly and trying to capture every single belief belief that leads to a research team whether they're in academia or not to carry out interpret analyze and communicate a certain topic that is a really far-reaching task however bayesian statistics and bayesian approaches like active inference are helping us get a grasp at the preconditioning that does play an essential role with ongoing experience so the question of how to engage with extra institutions and norms and

practices is a challenging and ongoing work but in some ways the formal principles that help us understand how context and measurement come together for cognitive entities and groups those fundamental principles are being

communicated maybe also i'd love to hear about what you're planning to talk about

yeah i'm not sure if i if i uh i'm not sure if i even fully understand the question um i took it the question just briefly i took it to be maybe um some worry about the the relationship between um the formalisms

and computational models inactive inference and human subjectivity itself so how do they how do they bridge the gap of phenomenology there um it's a really interesting question i know that there are some people who think that phenomenology is human subjective experience is just going to have these transcendental

aspects to it that can't be captured um with by the formalisms um that's something i won't be talking about in my session um what i will be talking about is um so some of the so i'm going to speak on some things that did come up briefly today um so many of the readings uh that feature in my section are uh readings that will use the term predictive processing so i was going to say a little bit quite briefly about how i think these different um different slightly different frameworks fit together and what some of the similarities and differences are i was going to say move quite quickly and superficially over some of the core mechanisms so precision weighting predictions generative models um and then i was the what i'm going to try and do is before we go on to the subsequent weeks where we look at collective behavior um norms normativity social scripts and things i wanted to try and build up a nicely layered picture of what it's like to be an active inference agent in the world and so the way i'm going to do this is by swooping over some philosophical territory so i'm going to look at the way active inference has um said interesting things about philosophy of mind particularly in terms of the role of internal representation

um if i have time i'm going to say a little bit about embodied cognition some of the four e stuff
um avil you got it right so it's embedded embodied and acted and extended and i'm not going to be able
to talk about all of those um but i'm going to say a little bit about some of them and how active
inference kind of brings interesting debates in that area out i'm going to say a little bit about folk
psychology and motivated action so talking briefly about um what agency itself is for an active influence
agent um and whether psychology folk psychology it's um it's a kind of propositional psychology so it's a
it's a framework of propositions that we generally use to describe and explain human behavior so it involves
uh propositions uh propositions such as a desires b or x hopes that y and it turns out that this propositional
psychology is pretty good um for explaining and predicting human behavior but it's not you say weeds um
20 slave persons philosophers

so do you mean we in the sense that it's it's a useful framework um i mean that you should explain
what the term is because uh it's full psychology is quite specific it's not a word that is in you know
folk psychology oh okay okay yeah so folk psychology is just basically the linguistic framework we use to
describe human behavior so if i say you know if i want to explain why um avell goes to the refrigerator
to grab a beer i would say you believe there's beer in the fridge you desire a beer and then on the basis
of that you form the intention to then go to the fridge um and it's been the subject of a lot of
philosophical interest because the the kind of ontological status of those concepts is very unclear so
it's not clear whether when we use folk psychological concepts are we saying that those those concepts
exist in some meaningful way or are we taking a more should we take a more pragmatic approach and
say that this is just instrumentally useful but we're not actually positing the existence of any kind
of brain states or anything like that um and there's been some interesting work on how predictive
processing and active inference might map onto folk psychology um are they in a necessary kind of
tension because obviously with predictive processing you have um just a very very limited set of primitives
with which to describe and explain human behavior so i'm going to say a little bit about that um and then
i'm going to move straight into talking about phenomenology and talking about how um certain aspects of
the framework have been said to underpin certain kinds of experiences so i'm going to use a case study of
addiction i think possibly depression but i think i'm going to talk about addiction and with this case
study i'm hoping to bring out um kind of tie together all of the things that i've said up to that point
and some of the things that you've covered as well and i should say as well in the discussion session
um for my section we'll have a special guest as well so we're going to be joined by professor mark miller
who is um a prolific author of predictive processing and neurophenomenology type stuff he's done some
really really interesting work on happiness and well-being depression addiction and really using the
active inference framework in some quite imaginative ways um and he's awesome in discussion so um
hopefully

you will find the readings for my section really interesting and if you do you'll have a chance to
um kind of have a conversation with the author for those pieces

avil any closing notes

i'm doing for the rules i would say

one short question i just want to get it on there while we're here brin asks

as a psychologist i am more or less in the business of helping people minimize their variational free energy and expected free energy great profession but what are the markers of free energy in the brain or organism is it heart rate neuromodulation galvanic skin response cortisol levels etc i say the uh closest proxy would be uh so those prediction error things in the brain uh so when we have expectations they are violated you have a bad preparation of this um information the fact the expectation was violated

and this would be the closest uh proxy to variational free energy um i do not think it would be the exclusive one but it is uh the main one and you could i don't think you could make an exhaustive list basically anything that is

uh out of order so as to prompt any kind of response of either you know dating models or factoring on things it would be a marker of uh well prediction error which is not quite a rational free energy but is also quite not not quite noted so for expected free energy um i have no idea whatsoever i don't think this framework is uh has made uh is connected explicitly to physiology yet uh am i wrong daniel it's very uh it's very interesting question and all i will say all i feel like i can say on it is um one thing we're going to talk about in my session is um how prediction error is not always a bad thing so there's a really nice a on article that came out fairly recently about the value of uncertainty and one of the things i'm going to talk

about is how um predictive processing has you know perhaps it can account for phenomenon such as uh playful

behavior creativity um engagement in challenging hobbies and dangerous sports and things like this and how we get an account of exploratory behavior and curiosity which is kind of rooted in the fact that for the creatures like we are for active inference agents um the resolution of prediction error itself is a good thing and so in some sense we're always seeking it out so it's i think it might be a mistake to kind of in it's very intuitive to equate prediction error of free energy with kind of allostatic load and stress and negativity but actually it's going to turn out that that's not the case yeah thank you and i'll give

another type of answer which is variational free energy and expected free energy are statistical calculations about statistical models so bodies don't have expected free energies models that are made about bodies can have variational and expected free energy and so to that end whatever you put in your model is either an observable or a proposed latent hidden state that you didn't measure and you may find that different experiential reports or different physiological phenomena have different relationships to each other so indeed it has been explored for example which neural firing rates are associated with surprise or which hormonal changes are associated with this or that change in body temperature when the generative model is made the deflationary answer is look at the data and find out which cues you measure are associated with your calculation of vfe and efe however i think it's a little bit of a map territory situation and one that's going to come up i am sure many times in our course

great well i'll just remind people to go to the course website to review the syllabus and also encourage everybody to submit questions through the course website all of the teachers will review the questions

and we'll give the best answers we can and also please from all backgrounds we would love to hear your voice in the discussions so thank you again ben avil see you next week bye
thank you